IA II-Activity 1 (20 Marks): Agentic AI System using Modern LLM-based Tools

Activity Type - Group Activity (3 students – *from same lab batch*)

Learning Objectives

After completing this activity, students will:

- Understand Agentic AI beyond classical agents
- Learn how LLMs act as reasoning agents
- Use tool-using, planning, and memory-based agents
- Compare classical AI agents vs modern agentic systems

Activity Description

Students will build a small Agentic AI system using modern AI tools where an LLM acts as an autonomous agent that:

- Understands a goal
- Plans steps
- Uses tools (search, calculator, documents)
- Produces an output

This activity exposes students to modern agentic AI systems using LLM-based frameworks, which extend classical agent concepts taught in the syllabus. The focus is on tool usage, reasoning, and orchestration rather than model training, ensuring feasibility for undergraduate students.

Al Tools

Students must use **at least ONE** of the following: (This is not the complete list, use something equivalent to the given list)

Agent Frameworks

- **AutoGen**
- **CrewAI**
- **ReAct-style agents**
- **BrowserUse**
- **Openrouter (free api)**
- **Langchain**

Tools integrated with agent

- Web search APIs
- Google OAuth 2.0 (Google Calender task)

- Calculator tools
- File/document readers
- Simple databases (CSV/JSON)
- Python tools

Example Problem Statements (Students choose ONE or their own idea)

1. **Research Assistant Agent**
    - Reads documents
    - Answers questions using tools
2. **Schedule meeting in google calendar**
    - Input prompt to schedule meeting
    - Automatically adds it to calendar
3. **Academic Planner Agent**
    - Plans study schedule
    - Adjusts based on constraints
4. **Healthcare Information Agent***(non-diagnostic)*
    - Collects symptoms
    - Retrieves medical guidelines
5. **Job Application Assistant Agent**
    - Analyzes resume + JD
    - Suggests improvements
6. **Travel Planning Agent**
    - Plans itinerary
    - Uses search + reasoning tools

Student Tasks

Part A – Agent Design (Modern View)

Students must explain:

- Goal of the agent
- Agent role
- Tools used
- Memory type (short-term / context)
- Planning strategy (ReAct / step-by-step)

Part B – Implementation (Modern AI Tools)

Minimum requirements:

- One **LLM-based agent**
- At least **two tools**

- Multi-step reasoning
- Observable agent behavior (logs / traces)

Example:

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User Goal → Agent Reasoning → Tool Call → Observation → Next Action → Final Answer
```

Part C – Analysis & Comparison

Students must **compare**:

Classical Agent (Syllabus)	Modern Agent (This Activity)
Rule-based	LLM-driven
Static logic	Dynamic planning
No memory	Contextual memory
Single-step	Multi-step reasoning

Deliverables

- PDF report (5–10 pages)
- Architecture diagram/Tool flow diagram
- Code repository / notebook + demo of working agent
- Agent interaction logs
- prompts and output screenshots

Evaluation Rubrics (20 Marks)

Component	Marks
Use of modern tools & frameworks	5
Agent reasoning & tool orchestration	5
Architecture & design clarity	5
Analysis & reflection	5